Ethan Bailie

+1(289) 221-0488 | eabailie@uwaterloo.ca | www.linkedin.com/in/ethan-bailie

Education

University of Waterloo Sept. 2018 – May 2023
Geomatics Honours Co-op, Computing Minor *Waterloo, Ontario, Canada*

- **Relevant Courses** | Data-Intensive Distributed Analytics, Data Types and Structures, Computer Applications: Databases, Spatial Databases, Algorithmic Problem Solving, Statistics I, Statistics II, Linear Algebra I

Experience

GIS Data Scientist January, 2022 – December 2022
TD Bank *Toronto, Ontario, Canada*

- Created data models using **big data ETL pipelines** by utilizing **SQL**, **Python**, and the ETL tool **Alteryx** to identify priority markets for growth and optimization
- Created ArcGIS **web applications** by analyzing demographic data as well as competitor location data to **highlight market opportunities** and prospective customers
- Accessed relevant data from **data marts** and **data warehouses** using **SQL** to **support existing projects** worked on both as an individual and as a team
- Created and **presented dashboards** to convey complex **analytical insights** to non-technical executives in a simple and understandable manner

GIS Data Analyst May, 2021 – August, 2021
Department of National Defence *Ottawa, Ontario, Canada*

- Wrote **Python scripts** to **automate** rebuilding a satellite imagery database that is used across the Department
- Analyzed satellite imagery and elevation data to **create simulations** of various locations to train military helicopter pilots
- Implemented **data standards** through writing documentation on best practices to **maintain imagery databases** and prevent further issues

GIS Data Analyst January, 2020 – April, 2020
JANA Corporation *Aurora, Ontario, Canada*

- Used **SQL** to search and filter through client databases for relevant data sets to be used for further processing
- Wrote **Python scripts** to **automate** complex geoprocessing tasks to assess client data and their associated risk
- Collaborated extensively with software and engineering teams through both formal meetings and informal conversations to **satisfy requirements, meet goals, and deliver on time**

IT Project Coordinator June 2019 – August, 2019
CIBC *Toronto, Ontario, Canada*

- Managed data privileges of existing employees through clear communication and **Excel** to **maintain database integrity**
- Created **presentations** for upper management by **collaborating** with other employees on various teams to convey to executives a better understanding of **project scope and progress**
- Updated key documents related to banking regulations through careful research and summarization of my findings

Technical Skills

Programming Languages: Python, R, SQL, Bash
Technologies: SQL Server, Excel, Looker Studio, Tableau, Airflow, PostgreSQL, Spark
Libraries/Frameworks: PySpark, ArcPy, Pandas, NumPy, MatPlotLib, RGDAL, TensorFlow, BNLearn

Projects

Bayesian Network Model | R (RGDAL, BNLearn)

- Implemented Bayesian network model to assess flood risk factors

Image Recognition Model | Python (TensorFlow, Pandas, NumPy, MatPlotLib)

- Created image recognition model that identifies different articles of clothing from still images, and produces statistics of guess probability